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Introduction

Medical devices do not manage themselves. From the moment a clinical need is identified through procurement, commissioning, training, maintenance and eventual decommissioning, every stage demands deliberate governance. Together, these stages form the medical equipment lifecycle. Fail to govern any one of them and the consequences are real: patient harm, staff injury, and institutional legal liability (MHRA, 2021).

This essay critically analyses the Management of Medical Devices Policy (Version 6, October 2024) published by Shropshire Community Health NHS Trust (SCHT). Serving approximately 455,000 people across 1,235 rural square miles from more than 130 service delivery locations, and relying on a single external engineering provider, SCHT faces governance challenges that expose every weakness in a device management policy (SCHT, 2024). The analysis addresses two questions: the roles and responsibilities of key personnel and groups, and how well the policy is likely to achieve its stated aims across the medical device lifecycle.

Roles and Responsibilities of Key Personnel and Groups

The SCHT policy creates a layered governance structure: authority flows downwards, accountability flows upwards, and three specialist functions operate laterally. This section examines each role and, where the policy fails to define performance metrics, proposes the measurable KPIs required to make governance robust across 130 dispersed locations. Table 1 maps the full structure, identifying the specific gap in each role and the KPI needed to close it.

Table 1: Roles, Accountability, Governance Gaps and Required KPIs for Key Personnel and Groups in the SCHT Management of Medical Devices Policy (Version 6, October 2024)

Role / Group	Accountability Direction	Key Responsibilities (SCHT, 2024)	Governance Gap or Accountability Weakness	Required KPI to Close Gap
Chief Executive	Accountable to NHS Trust Board. Delegates to Director of Governance.	Bears non-delegable legal responsibility for ensuring all device users are competent and all equipment is fit for purpose, maintained, and correctly used (CQC, 2014; Health and Safety at Work etc. Act 1974).	The policy assigns ultimate accountability to the Chief Executive but defines no board-level performance reporting frequency beyond the biannual Medical Devices Management Report. Systemic failure cannot be attributed to subordinate roles alone; without more frequent board visibility, the Chief Executive cannot discharge this legal duty proactively.	Annual completion of Medical Devices Management Report to the Board with zero overdue escalations. Biannual report presented on schedule with no outstanding unresolved Safety Alerts.
Director of Governance	Accountable to Chief	Oversees planning, operation, management,	The policy defines no performance metrics for	SLA formally reviewed annually

Critical Analysis of a Published Medical Devices Management Policy for SHT

Role / Group	Accountability Direction	Key Responsibilities (SHT, 2024)	Governance Gap or Accountability Weakness	Required KPI to Close Gap
	Executive. Delegates to Quality and Safety Committee and Health and Safety Committee.	and disposal of all medical devices across the Trust. Sponsors governance committees receiving device management reports.	this role, including no requirement to formally review the MES Service Level Agreement or evaluate whether MDMG governance structures are adequate for 130 dispersed sites. Without defined standards, deterioration in governance quality is invisible until an incident occurs.	by Director of Governance, with documented outcomes. Annual sign-off that MDMG membership, Terms of Reference, and meeting frequency remain adequate for the Trust's scale.
Quality and Safety Committee / Health and Safety Committee	Accountable to Director of Governance. Delegates to MDMG.	Receive biannual Medical Devices Management Report. Provide board-level assurance on device management standards and incident trends across the Trust.	A biannual reporting cycle means that serious device management deterioration may persist for up to six months before it reaches board-level awareness. The policy defines no mechanism for escalating urgent device safety issues between scheduled meetings, leaving a significant governance blind spot across 130 locations.	100% of National Patient Safety Alerts acknowledged within MHRA-mandated timescales. Urgent device safety issues escalated to committee within five working days of identification, evidenced by a documented escalation log.
Medical Devices Management Group (MDMG)	Accountable to Quality and Safety Committee. Delegates to Divisional Clinical Managers.	Monthly operational governance hub overseeing policy review, equipment approval, PPM scheduling, incident review, and training support. Multidisciplinary membership spanning clinicians, engineers, procurement staff, and device users.	Monthly meetings are the sole governance checkpoint for 130 service delivery locations. The policy defines no divisional sub-structure or escalation trigger between sites and the MDMG, meaning device management issues at remote clinics may not surface for up to four weeks. The MDMG thus relies entirely on voluntary upward reporting rather than systematic data collection (Temple-Bird et al., 2005).	PPM compliance rate reported monthly and maintained at or above 95%. Device incident review completed within one MDMG meeting cycle for 100% of reported incidents. Escalation trigger defined and documented for issues arising between monthly meetings.
Divisional Clinical	Accountable to MDMG.	Ensure staff within their division are trained and	The policy does not define how many	Monthly divisional Datix incident

Critical Analysis of a Published Medical Devices Management Policy for SCHK

Role / Group	Accountability Direction	Key Responsibilities (SCHK, 2024)	Governance Gap or Accountability Weakness	Required KPI to Close Gap
Managers	Delegates to Ward and Departmental Managers.	competent; areas are suitably equipped; device incidents and process failures are reported through Datix; remedial actions implemented where deficiencies are identified.	service locations each Divisional Clinical Manager oversees, nor what reporting mechanisms link divisional level to the MDMG. Without a defined reporting structure, the MDMG cannot confirm that divisional oversight is consistently functioning across the Trust.	summary submitted to MDMG. Training compliance audited at divisional level each quarter, with results reported to MDMG.
Ward and Departmental Managers (Budget Holders)	Accountable to Divisional Clinical Manager. Delegates to Medical Device Users.	Designated asset owners for all devices in their area. Maintain device inventory; agree staff authorization lists; establish local competency monitoring processes; ensure all devices display a current Next Test Due label.	The policy currently lacks any central mechanism to verify that local inventory management and competency monitoring are being maintained across 130 sites. Without central verification, standards may diverge silently from site to site, with no means of detection until a patient safety event occurs (Chan, 2003).	100% of devices within each area displaying a current Next Test Due label, verified by MES at each scheduled site visit. Device inventory reconciled against the central asset database quarterly and discrepancies reported to MDMG within five working days.
Medical Device Users	Accountable to Ward and Departmental Manager.	Pre-use safety checks; confirm service label currency; operate only devices for which trained; report faults and adverse events through Datix; comply with decontamination procedures (Health and Safety at Work etc. Act 1974).	The policy requires initial competency sign-off before device use but defines no minimum interval for refresher training. Without a mandated reassessment cycle, competency for high-risk devices such as infusion pumps and defibrillators may lapse silently across the Trust's 130 locations, particularly at remote sites where peer oversight is limited (Shields and Latter, 2019).	95% or more of users holding a current competency sign-off for each assigned device class at any point in time, verified quarterly through the central training register. 100% of device faults reported through Datix within 24 hours of identification.
Medical Device Safety Officer (MDSO)	Supports MDMG and Divisional Clinical Managers.	Coordinates response to National Patient Safety Alerts and Field Safety Notices; oversees Datix device incidents;	The policy does not define how the MDSO interfaces with Digital Services or the Clinical Safety Officer when a	100% of National Patient Safety Alerts acknowledged within mandated

Critical Analysis of a Published Medical Devices Management Policy for SCHAT

Role / Group	Accountability Direction	Key Responsibilities (SCHAT, 2024)	Governance Gap or Accountability Weakness	Required KPI to Close Gap
	Deputized by Health and Safety Advisor and CAS Liaison Officer.	determines whether MHRA Yellow Card reporting is required; manages continuity arrangements in their absence (MHRA, 2021).	connected device incident has both a clinical and a cybersecurity dimension. This information chain gap means that a connected device incident could be managed by either function independently, with no defined lead or escalation pathway.	timescale and response documented. MHRA Yellow Card submissions completed within 30 days of incident identification. Connected device incident protocol documented and tested annually with Digital Services.
Medical Engineering Services (MES / SaTH)	Delivers services under Service Level Agreement to MDMG and Ward Managers. Accountable to Director of Governance via MDMG.	Maintains live asset database; acceptance testing per MHRA Tables 5.1 and 5.2; schedules and delivers PPM; conducts device risk assessments; manages data erasure at end of life (SCHAT, 2024).	The policy defines no SLA performance standards for MES, no maximum repair turnaround time, no minimum PPM completion rate, no uptime requirement, and no contingency if MES cannot perform across the 130 sites it services. This means that any deterioration in engineering service quality is invisible to SCHAT governance until patients are already at risk. MES is also not required to demonstrate ISO 13485:2016 accreditation (Temple-Bird et al., 2005).	PPM completion rate of 95% or above reported to MDMG monthly. Emergency repair response within five working days for 100% of reported faults. SLA formally reviewed annually by Director of Governance, with written outcome report. ISO 13485:2016 accreditation certificate provided to Director of Governance annually.
Digital Services	Supports MDMG and MES. Accountable to Director of Governance.	Cybersecurity risk assessments before any connected device purchase; approval before connected devices join the Trust network; ensures reliable data transmission; supports data erasure for connected devices at end of life (SCHAT, 2024; NHS Digital, 2022).	The policy does not define a maximum timeline for applying software security patches to connected medical devices. Without a mandated patch window, known vulnerabilities may remain open for extended periods across networked devices at all 130 sites, replicating the conditions that enabled the WannaCry attack to disable NHS clinical systems in 2017 (Ghafur et al., 2019).	100% of critical security patches applied to connected medical devices within 30 days of release. Patch compliance rate reported to MDMG monthly. Cybersecurity incident response plan for connected devices exercised annually with documented outcomes.

Critical Analysis of a Published Medical Devices Management Policy for SHT

Role / Group	Accountability Direction	Key Responsibilities (SHT, 2024)	Governance Gap or Accountability Weakness	Required KPI to Close Gap
Clinical Safety Officer (CSO)	Accountable to Director of Governance. Works alongside Digital Services.	Manages clinical risk from new digital systems and connected medical devices; reviews and signs off clinical safety cases for new digital device deployments (SHT, 2024).	The policy does not specify at what stage of procurement or deployment CSO sign-off is required, nor how CSO findings are escalated to the MDMG. Without a defined entry point in the procurement process, a connected device could theoretically be deployed clinically before its clinical safety case is formally assessed.	CSO sign-off documented for 100% of new digital device deployments prior to first clinical use. CSO findings escalated to MDMG within five working days where clinical risk is identified. Escalation log reviewed at each MDMG meeting.

Adapted from SHT (2024, pp. 5–10). MDMG = Medical Devices Management Group; MDSO = Medical Device Safety Officer; MES = Medical Engineering Services; SaTH = Shropshire and Telford Hospitals NHS Trust; CSO = Clinical Safety Officer; PPM = Planned Preventive Maintenance; SLA = Service Level Agreement.

Governance Hierarchy

The Chief Executive carries non-delegable legal responsibility for the entire device management system. Grounded in Regulation 15 of the Health and Social Care Act 2008 (Regulated Activities) Regulations 2014, this duty means that even when operational functions are delegated, personal accountability for systemic failure stays at board level (CQC, 2014). There is nowhere to pass it.

Operational responsibility is delegated to the Director of Governance, before flowing through the Quality and Safety Committee and Health and Safety Committee and reaching the Medical Devices Management Group (MDMG), which meets monthly and functions as the operational governance hub, with multidisciplinary membership spanning clinicians, engineers, procurement staff, and device users. Weng et al. (2022) found that ward-based multidisciplinary team structures had a significant positive effect on patient safety culture, demonstrating that collective governance produces better safety outcomes than siloed professional working.

Yet monthly MDMG meetings are the only governance checkpoint for 130 locations. A device problem at a remote clinic may wait up to four weeks before it reaches governance level. Temple-Bird et al. (2005) are clear: dispersed organizations require intermediate reporting layers between sites and the central governance body. The SHT policy defines neither intermediate layers nor escalation triggers, leaving the MDMG entirely dependent on staff choosing to report upwards rather than being required to.

Below the MDMG, Divisional Clinical Managers drive training and incident reporting across their division. Ward and Departmental Managers own the assets in their area: inventories, staff authorization lists, and local competency records. Placing accountability at the point of

care makes clinical sense (Chan, 2003). The problem is verification. With 130 locations and no central monitoring mechanism, there is no way to confirm that any of this is actually happening. Chan (2003) warns that without oversight, device management quality diverges across sites. In a Trust of this scale, divergence is not a risk to be managed; it is an outcome to be expected unless the system actively prevents it.

Individual Medical Device Users carry delegated responsibility for pre-use checks, confirming service labels are current, using only devices for which they are trained, and reporting faults through Datix (SCHAT, 2024). The Health and Safety at Work etc. Act 1974 imposes a parallel personal legal duty of care on users, confirming that training obligations protect staff as well as patients.

Specialist Support Roles

Three specialist functions operate laterally. The Medical Device Safety Officer (MDSO) is specifically recommended by the MHRA (2021) for all healthcare organizations. The MDSO coordinates responses to National Patient Safety Alerts and Field Safety Notices, and oversees Datix incidents to confirm MHRA Yellow Card reporting is completed appropriately (SCHAT, 2024). Continuity is provided by the Health and Safety Advisor and CAS Liaison Officer, who deputize in the MDSO's absence.

Technical services are delivered entirely by Medical Engineering Services (MES), provided by Shropshire and Telford Hospitals NHS Trust under a Service Level Agreement. MES covers everything: the asset database, acceptance testing, PPM scheduling, risk assessments, and data erasure (SCHAT, 2024). That concentration of function in a single external provider, serving 130 dispersed sites, is a structural vulnerability the policy does not acknowledge. No SLA performance standards exist: no repair turnaround target, no PPM completion floor, no contingency if MES cannot deliver. Temple-Bird et al. (2005) are explicit that contracts of this kind demand defined performance metrics and fallback provisions, especially where there is no in-house engineering capacity. Without them, SCHAT cannot know whether MES is performing adequately until a patient is already at risk.

Version 6 adds Digital Services and a Clinical Safety Officer (CSO) to the framework. Digital Services is accountable for cybersecurity risk assessments and ensuring reliable data transmission from connected devices (SCHAT, 2024; NHS Digital, 2022). The CSO manages clinical risk from new digital systems. These additions are necessary and timely. However, the policy does not define how Digital Services and MES coordinate when a connected device has both an engineering and a cybersecurity dimension, creating an interface gap that could delay the clinical response.

In summary, the governance structure is logically layered and covers the right personnel. Its three critical weaknesses are: no SLA performance standards or contingency for MES; no central compliance monitoring across 130 locations; and an undefined interface between MES, Digital Services, and the MDSO for connected device incidents.

Critical Analysis of the SCHAT Policy Against the Medical Device Lifecycle

The medical equipment lifecycle offers the most rigorous framework for evaluating a device management policy. Temple-Bird et al. (2005) and WHO (2011) describe seven phases, each driven by a Technology Needs Assessment: Planning and Acquisition, Acceptance, Training, Maintenance, Quality Assurance, Risk Management, and Replacement and Disposal (see Figure 1). Weakness in any phase generates risk that compounds across the device's entire working life. The SCHAT policy sets out an ambitious aim: to equip the Trust with innovative, secure, connected devices that improve care quality and ensure long-term sustainability (SCHAT, 2024, p. 4). The question is whether the policy's provisions can actually deliver that.

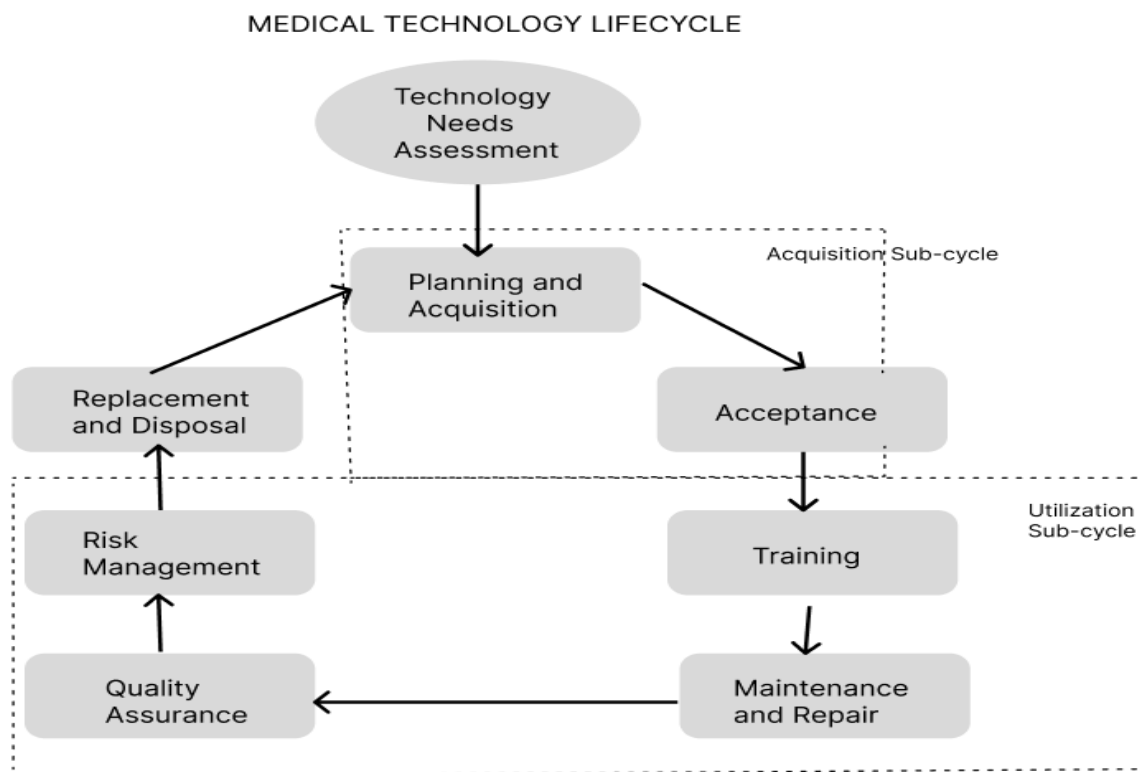


Figure 1: Medical Equipment Lifecycle. Adapted from Temple-Bird et al. (2005) and WHO (2011).

The lifecycle comprises seven operational phases driven by a Technology Needs Assessment, organized into an Acquisition Sub-cycle (Phases 1–2) and a Utilization Sub-cycle (Phases 3–6), with Phase 7 closing the cycle and feeding back into the next Technology Needs Assessment.

Phase 1: Planning and Acquisition

Poor decisions in Phase 1 cannot be undone downstream. The SCHAT policy establishes sound procurement controls: approved suppliers only, MDMG sign-off for exceptions, and Capital Management Group approval for purchases above £5,000 (SCHAT, 2024). That structure is appropriate. What the policy fails to mandate, however, is a structured Technology Needs Assessment before any purchase, despite the MHRA (2021) specifically requiring one. Across 130 sites with varied clinical environments, a device procured without systematic assessment of compatibility, lifetime cost, and decontamination requirements will create

training and maintenance problems that cascade across the Trust. The policy also offers no guidance on donated or loaned devices beyond requiring acceptance testing, a significant omission given that IPEM (2024) classifies these as higher risk precisely because their service history and residual life are unknown.

Phase 2: Acceptance

Phase 2 is where the policy performs best. Every new device must be commissioned jointly by a MES biomedical engineer and a Trust staff member before clinical use, covering delivery verification, functional acceptance testing per MHRA Tables 5.1 and 5.2, safety labelling, asset registration, and risk assessment (SCHAT, 2024; MHRA, 2021). Version 6 strengthens this further by requiring Digital Services notification before any connected device joins the Trust network, this is a necessary addition given the sector-wide vulnerability to cyber threats (NHS Digital, 2022).

However, the policy states only that “adequate user verification” must be carried out for in vitro diagnostic (IVD) devices, blood glucose monitors, urine analyzers, and C-reactive protein testing equipment that generate results used directly in treatment decisions, without defining what adequate verification means in practice (SCHAT, 2024, p. 15). Fanshawe et al. (2020), in a mixed-methods evaluation of point-of-care testing platforms across a large acute NHS trust, found that pre-analytical errors from operator handling failures were a persistent source of inaccurate results, with inadequate acceptance testing identified as a key contributing factor. In a community setting where IVD results determine whether a patient requires emergency transfer or antibiotic prescription, an inaccurate result from a poorly commissioned device carries immediate patient safety consequences. The Medical Devices Regulations 2002 (SI 2002/618, as amended) impose specific requirements for IVD acceptance that the SCHAT policy does not reference; a legal compliance gap requiring priority attention.

Phase 3: Training

Inadequate training remains one of the MHRA’s most frequently identified causes of device-related harm (MHRA, 2021), and the risk is sharper in a community trust where staff at remote sites lack the peer oversight that a hospital ward provides. The Health and Safety at Work etc. Act 1974 places a legal duty on employers to protect both staff and patients. The SCHAT policy addresses training content well: operation, alarm management, decontamination, fault recognition, and incident reporting are all required, and obligations extend to patients and carers receiving home-use devices (SCHAT, 2024). Training that is properly structured works: Shields and Latter (2019) showed a competency jump from 65% to 97% following a structured ward-based program.

Three gaps undermine this otherwise solid framework. First, the policy defines no minimum refresher interval. Infusion pump competency that lapses silently can lead to tenfold medication overdoses; outdated defibrillator training costs lives (MHRA, 2021). Shields and Latter (2019) name the absence of mandated intervals as a recurring NHS failure: without a deadline, refresher training happens when convenient, not when needed. Second, all devices face identical training requirements regardless of risk. A risk-stratified approach would direct

intensive schedules toward high-risk devices, improving safety and efficiency simultaneously (MHRA, 2021). Third, training records are held only at departmental level. There is no Trust-wide view of compliance across 130 sites. Taghipour et al. (2011) show that centralized monitoring is essential for catching systemic gaps before they reach patients.

The absence of centralised monitoring is not merely an administrative inconvenience; it is a structural incompatibility between the policy's governance ambitions and the operational reality of a 130-site community trust. Within a single hospital, a ward manager can observe whether colleagues are completing refresher training and escalate concerns to a clinical educator. At a remote SHT clinic, that peer oversight does not exist. Training compliance is self-reported upwards through Divisional Clinical Managers to the MDMG, a chain that depends entirely on voluntary disclosure of non-compliance. There is no mechanism by which the MDMG can independently confirm that the training rates being reported reflect reality on the ground.

Taghipour et al. (2011) propose a centralised equipment management information system in which training compliance data, maintenance records, and device incident rates are integrated into a single dashboard visible to the governance body. Applied to SHT, this would mean every Ward and Departmental Manager submitting training completion data to a central register accessible to the MDMG, rather than holding it locally at departmental level. The MDMG could then identify sites with declining compliance before incidents occur, rather than discovering the gap through a Datix report after one has. This is not a novel or disproportionate requirement: NHS Golden Jubilee (2022) mandates precisely this kind of centralised training record as part of its lifecycle management framework.

The policy's current arrangement inverts the logic of governance. It places accountability at the point of care, which is correct, but then provides no means of verifying whether that accountability is being discharged. Accountability without verification is not governance, it is delegation without oversight. In a Trust operating across 130 dispersed sites with no in-house engineering fallback and a single external provider, the margin for undetected compliance failure is one that SHT cannot afford.

Phase 4: Maintenance and Repair

The PPM program is a genuine strength: annual schedule, monthly MES site visits, manufacturer-compliant servicing, and a five-day repair target where parts are available (SHT, 2024; MHRA, 2021). However, the policy fails to answer a critical question: when MES time is limited across 130 sites, which device gets serviced first? Without a prioritization methodology, the answer is effectively whoever is next on the schedule. Chan (2003) proposes a four-criterion scoring index, Safety, Performance, Availability, and Maintenance Requirement, that directs engineering effort toward devices where failure carries the greatest clinical harm. At SHT, where the only infusion pump or defibrillator at a remote clinic may be the device awaiting service, this gap is not theoretical: it is a direct patient safety risk. Taghipour et al. (2011) validated a related approach and found it significantly reduced unplanned downtime for high-consequence devices. The policy also ignores Reliability-Centred Maintenance (RCM), the evidence-based approach in which

maintenance strategy is designed around each device's specific failure modes and their clinical consequences (IPEM, 2024). A uniform schedule applied equally to all device types will inevitably miss the most dangerous failure modes in some of them.

RCM, developed in the aviation industry and subsequently validated in healthcare engineering contexts, proceeds from a fundamentally different premise than a uniform schedule. Rather than asking "when should this device be serviced?", RCM asks "what are the specific ways this device can fail, what is the clinical consequence of each failure mode, and what maintenance intervention most effectively prevents that consequence?" (IPEM, 2024). The answers differ substantially between device classes. A defibrillator's most clinically dangerous failure mode is capacitor degradation reducing delivered energy which is time-dependent and requires scheduled energy verification testing at defined intervals regardless of use frequency. An infusion pump's most dangerous failure mode is motor wear causing flow rate inaccuracy that is use-dependent and requires maintenance intervals calibrated to cumulative operating hours rather than calendar time. A uniform annual PPM schedule applied to both devices will be simultaneously over-engineered for some failure modes and dangerously under-engineered for others.

At SCHAT, where the defibrillator at a remote rural clinic may be the only one within a clinically significant response radius, the consequence of a maintenance schedule that does not distinguish between device criticality and failure mode profile is not theoretical. It is a direct, foreseeable patient safety risk of the kind that ISO 14971:2019 requires organisations to identify and control (ISO, 2019). Incorporating RCM principles into the MES contract, even at a simplified, risk-stratified level, would represent a material improvement in the clinical safety of the Trust's maintenance program without requiring substantial additional resource.

Phase 5: Quality Assurance

The oversight structure is sound: device management sits on Divisional Quality and Safety meeting agendas, the MDMG receives bimonthly MES data, and the Health and Safety Committee receives biannual board-level reports (SCHAT, 2024). However, the policy fails to mandate ISO 13485:2016 accreditation for MES, the international quality management standard for device servicing organizations (ISO, 2016). ISO 13485 certification does something internal reviews cannot: it brings an accredited third-party body to independently verify that MES processes are documented, consistent, and improving. The MHRA (2021) recommends ISO 13485-compliant providers as best practice. NHS Golden Jubilee (2022) goes further and requires annual independent audits. The SCHAT policy says nothing about whether MES holds this accreditation. For an organization with no in-house engineering capacity and 130 sites entirely dependent on a single provider, that silence is a serious transparency failure.

Phase 6: Risk Management

Risk Management, as defined under ISO 14971:2019, encompasses systematic hazard identification, estimation of the probability and severity of harm, and implementation of proportionate controls across the device lifecycle (ISO, 2019). The SCHAT policy operates at

two levels. Proactively, MES risk-assesses all registered devices and enters high-risk devices onto the Trust risk register. Reactively, a clear sequential incident protocol is defined: preserve safety; quarantine and label the device; record settings and circumstances; notify the MDSO; preserve evidence; and trigger a MES investigation report attached to the Datix record (SCHAT, 2024). This reactive framework is consistent with WHO (2021) adverse event management guidance.

The policy's risk management provisions are thorough but reactive: they respond to device failures, not systemic governance gaps. Applying the ISO 14971:2019 framework proactively to those gaps tells a different story, one of multiple unmitigated risks with direct patient harm potential, made worse by the fact that SCHAT's 130 remote locations have no backup equipment and no in-house engineering response when MES cannot attend. Table 2 structures that analysis: eight critical gaps, each rated for likelihood and severity, each with its residual risk and a specific control and KPI required to close it.

Table 2: Proactive Risk Analysis of Key Policy Gaps: SCHAT Management of Medical Devices Policy (Version 6, October 2024)

Policy Gap / Hazard	Likelihood (1–5)	Severity (1–5)	Risk Level	Current Policy Control	Residual Risk	Recommended Control and KPI
No cybersecurity incident response plan for connected devices (e.g. ransomware disabling infusion pumps at remote sites with no backup)	3	5	Critical	Version 6 adds risk assessments and asset register cyber risk recording (SCHAT, 2024). No active response procedure defined.	High: no procedure to isolate devices, recover systems, or clinically substitute if networked equipment is disabled across multiple remote sites.	Develop and annually exercise a device-specific cybersecurity incident response plan. KPI: plan exercised once per year; documented recovery time objective of four hours or less for life-critical devices.

Critical Analysis of a Published Medical Devices Management Policy for SCHAT

Policy Gap / Hazard	Likelihood (1–5)	Severity (1–5)	Risk Level	Current Policy Control	Residual Risk	Recommended Control and KPI
No defined timeline for applying software security patches to connected medical devices	4	4	Critical	Digital Services must approve connected devices before network access (SCHAT, 2024). No patch schedule defined.	High: known vulnerabilities remain exploitable for extended periods; conditions similar to those preceding the WannaCry attack (Ghafur et al., 2019; Nurse et al., 2017).	Define maximum patch deployment window for all connected devices. KPI: 100% of critical patches applied within 30 days of release, reported monthly to MDMG.
No maintenance prioritization methodology across 130 sites, a single device at a remote clinic may be the only one available and receives no priority	3	5	Critical	PPM program with monthly MES site visits (SCHAT, 2024). No criticality scoring to direct engineering effort when resources are limited.	High: engineering time distributed by schedule rather than clinical consequence; high-priority devices at remote sites not distinguished from lower-risk equipment.	Implement four-criterion prioritization scoring (Chan, 2003): Safety, Performance, Availability, Maintenance Requirement. KPI: criticality index documented for every device; high-scoring devices serviced first within each maintenance cycle.
No proactive replacement planning, device failure at a remote clinic leaves patients without equipment until MES can respond	3	4	High	Reactive decommissioning criteria and Condemnation Certificate process defined (SCHAT, 2024). No forward-looking replacement index.	Medium-High: replacement triggered by failure rather than approaching end of life; remote locations disproportionately affected due to absence of backup	Implement rolling replacement priority index combining device age, maintenance cost, failure rate, and clinical importance (Chan, 2003). KPI:

Critical Analysis of a Published Medical Devices Management Policy for SCHAT

Policy Gap / Hazard	Likelihood (1–5)	Severity (1–5)	Risk Level	Current Policy Control	Residual Risk	Recommended Control and KPI
					equipment.	replacement plan reviewed annually; no device used beyond its risk-scored replacement date.
No minimum refresher training interval for high-risk devices, competency may lapse silently, particularly at remote sites	4	4	High	Initial competency sign-off required before device use (SCHAT, 2024). No minimum reassessment interval defined.	Medium-High: lapsed competency undetected until incident; amplified at remote sites where peer oversight is limited (Shields and Latter, 2019).	Define minimum refresher intervals by device risk class (e.g. annually for Class IIb devices). KPI: 95% or more of users with current refresher sign-off at any time, verified quarterly through central training register.
No SLA performance standards for MES, deterioration in engineering service quality is invisible to SCHAT governance	3	4	High	MES performance data reported to MDMG bimonthly (SCHAT, 2024). No performance floor defined in the SLA.	Medium: any deterioration in repair turnaround or PPM completion is only identifiable retrospectively, after clinical risk has already materialized.	Define minimum SLA standards: PPM completion 95% or above, emergency repair within five working days (100%). KPI: monthly MES performance dashboard to MDMG; SLA breach triggers formal review within five

Critical Analysis of a Published Medical Devices Management Policy for SCHAT

Policy Gap / Hazard	Likelihood (1–5)	Severity (1–5)	Risk Level	Current Policy Control	Residual Risk	Recommended Control and KPI
						working days.
No incident reporting culture provisions, under-reporting masks the true device risk picture across 130 sites	4	3	High	Datix reporting required; MDSO oversees incident reports (SCHAT, 2024). No provisions to encourage or incentivize reporting.	Medium: widespread under-reporting (NHS England and MHRA, 2014) means MDMG risk picture is incomplete; systemic device hazards may go undetected.	Introduce anonymized incident learning reviews at each MDMG meeting; provide visible staff feedback on actions taken. KPI: device incident reporting rate per 1,000 device uses tracked quarterly; year-on-year improvement required.
MES not required to hold ISO 13485:2016 accreditation, quality of engineering services cannot be independently verified	2	4	Medium	MES performance reviewed bimonthly by MDMG (SCHAT, 2024). No requirement for independent third-party certification of service quality.	Medium: quality of engineering processes dependent on internal SaTH governance rather than an externally audited management system (ISO, 2016).	Require MES to demonstrate ISO 13485:2016 certification. KPI: accreditation certificate reviewed annually by Director of Governance as part of SLA review; lapse in certification treated as SLA breach.

Risk assessed in accordance with ISO 14971:2019 (ISO, 2019). Likelihood and Severity rated 1 (lowest) to 5 (highest). Risk Level = Likelihood × Severity: Critical ≥15; High 9–14; Medium 4–8. Sources: Chan (2003); Ghafur et al. (2019); ISO (2019); NHS England and MHRA (2014); Nurse et al. (2017); SCHAT (2024); Shields and Latter (2019); Temple-Bird et al. (2005).

A further gap concerns incident under-reporting. A Patient Safety Alert issued jointly by NHS England and MHRA in 2014 directed trusts to actively improve device incident reporting rates, having found under-reporting to be widespread (NHS England and MHRA, 2014). National data suggest device incidents represent approximately 2% of all reported

patient safety events, a figure almost certainly below the true rate given by known barriers including time pressure and fear of blame (NHS England, 2024). The NHS Patient Safety Strategy (NHS England, 2019) commits organizations to developing a culture of open, blame-free reporting. The SCHAT policy contains no provisions for building such a culture, no team-based incident learning reviews, no staff feedback mechanism, and no metric for tracking whether reporting rates improve over time.

This gap is structurally amplified by SCHAT's geography. Under-reporting in NHS organisations is partly attributable to time pressure and fear of blame, but it is also driven by the absence of a visible reporting culture makes staff less likely to report when they never see evidence that reporting leads to action (NHS England, 2019). In a busy acute hospital, a near-miss involving a device generates visible local discussion, a visible Datix response, and visible feedback to staff. At a remote SCHAT community clinic staffed by a small team operating with limited management presence, none of those social and organisational cues exist. A device fault may be resolved informally, never entered into Datix, and never reach the MDMG. The result is that the MDMG's risk picture for remote sites is systematically less complete than for locations with stronger peer oversight, precisely the sites where the consequences of undetected device failure are most severe, because no backup equipment exists and MES response times are longest. Introducing anonymised incident learning summaries as a standing agenda item at MDMG meetings, with written feedback circulated to all sites, would address this structural imbalance at minimal organisational cost.

Phase 7: Replacement and Disposal

The SCHAT policy defines clear criteria for removing a device from service: worn out beyond economic repair, clinically or technically obsolete, lacking spare parts, subject to manufacturer recall, or superseded by a more effective alternative (SCHAT, 2024). MES issues a formal Condemnation Certificate; the device is removed from the active register while service history is retained; disposal must comply with the Waste Electrical and Electronic Equipment (WEEE) Regulations; and patient-identifiable data must be securely erased before disposal (SCHAT, 2024). The MDMG monitors condemned items for trends through a monthly MES report, which can identify emerging reliability problems with specific device models.

The decommissioning criteria are clear, but the policy stops short of asking the more important question: how does SCHAT identify a device that is about to fail before it actually does? No proactive replacement planning framework exists. When a device fails at a remote clinic, patients may have no alternative equipment until MES can respond. Chan (2003) proposes a replacement prioritization index combining device age, maintenance cost history, parts availability, clinical importance, and failure frequency into a rolling program that replaces devices before they become clinical incidents. Without it, SCHAT will keep reacting to failures rather than preventing them, and the locations that will suffer most are the remote ones with no backup (Temple-Bird et al., 2005). Table 3 provides a summary assessment of the policy across all seven lifecycle phases.

Table 3: *Summary Assessment of the SCHAT Management of Medical Devices Policy (Version 6, October 2024) Against the Seven-Phase Medical Equipment Lifecycle*

Critical Analysis of a Published Medical Devices Management Policy for SCHAT

Lifecycle Phase	Rating	Policy Strength	Critical Gap	Priority
1. Planning and Acquisition	Moderate	Two-tier financial control (SCHAT, 2024); MDMG approval for unapproved suppliers	No Technology Needs Assessment checklist (MHRA, 2021); no guidance on donated or loaned devices (IPEM, 2024)	High
2. Acceptance	Strong	Commissioning per MHRA Tables 5.1–5.2 (MHRA, 2021); digital notification added in Version 6 (NHS Digital, 2022)	No IVD-specific acceptance protocol; Medical Devices Regulations 2002 (SI 2002/618) not referenced (Fanshawe et al., 2020)	Medium
3. Training and Education	Moderate	Thorough training content; extends to patients and carers (SCHAT, 2024; Shields and Latter, 2019)	No minimum refresher interval; no risk-stratified training; no central compliance monitoring across 130 sites (Taghipour et al., 2011)	High
4. Maintenance and Repair	Moderate	PPM program monthly site visits; 5-day repair target; Next Test Due labels (SCHAT, 2024; MHRA, 2021)	No prioritization methodology (Chan, 2003); no RCM framework (IPEM, 2024); no SLA performance standards (Temple-Bird et al., 2005)	High
5. Quality Assurance	Moderate	Multi-level oversight; biannual board report; MDMG incident review (SCHAT, 2024)	No ISO 13485 accreditation requirement for MES (ISO, 2016; NHS Golden Jubilee, 2022); no defined audit frequency	Medium
6. Risk Management	Strong	Proactive device risk register; sequential incident protocol; Yellow Card reporting; Version 6 cybersecurity additions (SCHAT, 2024)	No reporting culture provisions (NHS England, 2019); no patch timeline; no cybersecurity incident response plan. See Table 2 for full risk analysis (Ghafur et al., 2019; Nurse et al., 2017)	High
7. Replacement and Disposal	Moderate	Clear decommissioning criteria; Condemnation Certificate; WEEE and data erasure compliance (SCHAT, 2024)	No proactive replacement priority index; replacement remains reactive and crisis-driven (Chan, 2003; Temple-Bird et al., 2005)	Medium

Based on Temple-Bird et al. (2005) and WHO (2011). Sources cited within cells. Rating: Strong = meets current best practice; Moderate = adequate basic provision but significant gaps identified.

Conclusion

The SCHAT Management of Medical Devices Policy (Version 6, October 2024) provides a governance framework grounded in MHRA guidance and UK legal requirements that addresses the complete medical device lifecycle. Its layered accountability structure is logically designed, its acceptance testing provisions are thorough, and the incorporation of connected device governance and cybersecurity roles in Version 6 is a necessary and timely

evolution. If consistently applied, the policy provides a sound foundation for patient and staff safety.

Yet significant gaps remain, and their clinical consequences fall hardest on SCHAT's most vulnerable points: 130 dispersed locations, no in-house engineering, and no backup equipment at remote sites. The two most urgent failures demand immediate action. The policy contains no cybersecurity incident response plan for connected devices. A ransomware attack disabling networked infusion pumps across multiple remote clinics simultaneously, the scenario WannaCry demonstrated at national scale (Ghafur et al., 2019), would leave clinical teams without either functional equipment or a written response procedure. The required KPI: an exercised response plan with a four-hour or less recovery time objective for life-critical devices. Equally urgent, the policy offers no maintenance prioritization methodology. Without criticality scoring, a defibrillator at an isolated rural clinic waits its turn in the same queue as a low-risk syringe driver at a busy hospital site. The required KPI: a monthly MES dashboard reporting PPM completion by device criticality tier, maintained at 95% or above. Beyond these, the remaining gaps identified in Tables 1 and 2, from the absent Technology Needs Assessment checklist and IVD compliance gap to the missing SLA standards and replacement index, should be addressed in the next scheduled policy revision.

The CQC found SCHAT's governance systems required improvement in 2016 (CQC, 2016). Version 6 represents genuine progress. But progress is not the same as adequacy. The gaps identified here are not marginal. They are the difference between a policy that documents intentions and one that prevents harm. Closing them, urgently for the highest-risk failures, systematically for the rest, is what would make the Trust's ambition of innovative, secure, and sustainable connected medical device management a governance reality rather than a stated aim.

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